Maths Clinic Week 1 Nov 29th, 30th & Dec 1st.

Matrices, Graphs & Trees - Jan 2023.

Question 1

a) $A = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$ $B = \begin{bmatrix} 5 & 6 & 0 \\ -3 & 1 & 8 \end{bmatrix}$ $C = \begin{bmatrix} 1 & -2 & 0 \\ 5 & 1 & 2 \end{bmatrix}$

i) size of Matrix A and B?

size = No. of rows × No. of columns

$A = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$ row 1, row 2, row 3; col 1

size = r × c = 3 × 1

$B = \begin{bmatrix} 5 & 6 & 0 \\ -3 & 1 & 8 \end{bmatrix}$ row 1, row 2; col 1, col 2, col 3

size = r × c = 2 × 3

ii) elements a_{ij}

Row, column.

∴ b_{22} is element in matrix B in row 2 col 2

$B = \begin{bmatrix} 5 & 6 & 0 \\ -3 & 1 & 8 \end{bmatrix}$ r2, c2

$b_{22} = 1$

a_{11} → element in Matrix A in row 1 Col 1

$$A = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix} \quad r1$$

Col 1

$a_{11} = 3$.

iii) Find the Matrix given by $B - 3C$.

get $3C$ i.e. multiply Matrix C by the scalar (3)

$$3\begin{bmatrix} 1 & -2 & 0 \\ 5 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 3(1) & 3(-2) & 3(0) \\ 3(5) & 3(1) & 3(2) \end{bmatrix}$$

multiply each element in the Matrix by 3.

$$3C = \begin{bmatrix} 3 & -6 & 0 \\ 15 & 3 & 6 \end{bmatrix}$$

$B - 3C$

$$\begin{bmatrix} 5 & 6 & 0 \\ -3 & 1 & 8 \end{bmatrix} - \begin{bmatrix} 3 & -6 & 0 \\ 15 & 3 & 6 \end{bmatrix} = \begin{bmatrix} 5-3 & 6-(-6) & 0-0 \\ -3-15 & 1-3 & 8-6 \end{bmatrix}$$

subtract corresponding elements from each other

$$B - 3C = \begin{bmatrix} 2 & 12 & 0 \\ -18 & -2 & 2 \end{bmatrix}$$

b) the determinant of matrix A (3×3 matrix)

Note: if only asked for the determinant of 3×3, use shorter method (by diagonals) - simpler method.

Repeat the first 2 cols.

$$A = \begin{matrix} 2 & -1 & 1 & 2 & -1 \\ 4 & -2 & 3 & 4 & -2 \\ 4 & -1 & 5 & 4 & -1 \end{matrix}$$

start at top left corner take each diagonal

$$\det A = (2\times -2\times 5) + (-1\times 3\times 4) + (1\times 4\times -1) - (4\times -2\times 1) + (-1\times 3\times 2) + (5\times 4\times -1)$$

determinant

start bottom left corner take each diagonal

$$= (-20) + (-12) + (-4) - (-8) + (-6) + (-20)$$

$$\det A = -36 - (-34) = \boxed{-2}$$

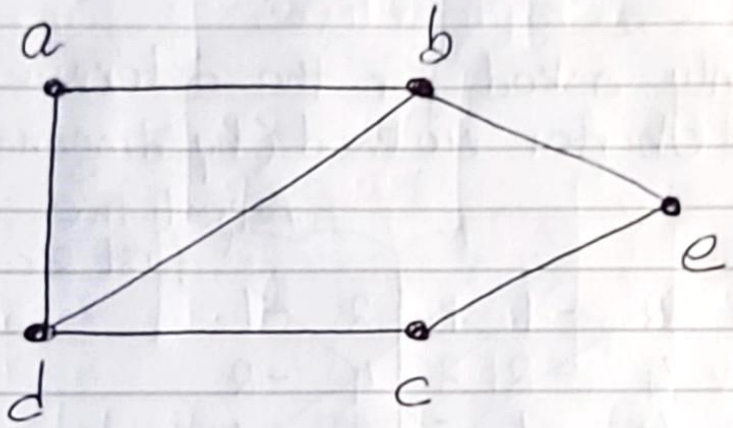
d) Draw the graph G for the given adjacency matrix:

	a	b	c	d	e
a	0	1	0	1	0
b	1	0	0	1	1
c	0	0	0	1	1
d	1	1	1	0	0
e	0	1	1	0	0

← labelling the vertices.

5 Rows & 5 Cols ⇒ 5 vertices in graph

graph G



your graph may look different depending on where you located each vertex. what matters is that the adjacencies are correct.

The degree of each vertex is the number of edges incident to (touching) the vertex. For the graph above

degree (A) = 2
degree (B) = 3
degree (C) = 2
degree (D) = 3
degree (E) = 2

Question 4: Matrices, Graphs & Trees.

a) $A = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 9 & 10 \\ 2 & 5 \end{bmatrix}$

i) Identity Matrix of B.

B is a 2×2 matrix therefore looking for the 2×2 Identity matrix

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

ones on the diagonal
zeros everywhere else.

Identity matrix always denoted/called I

ii) Evaluate $(-2B^T + AB)$

B^T ~ Transpose $\Rightarrow B^T = \begin{bmatrix} 9 & 2 \\ 10 & 5 \end{bmatrix}$

i.e. first row of matrix B becomes first col in B^T
second row of Matrix B becomes second col in B^T

$$-2B^T = -2\begin{bmatrix} 9 & 2 \\ 10 & 5 \end{bmatrix} = \begin{bmatrix} -18 & -4 \\ -20 & -10 \end{bmatrix}$$

Matrix multiplication

$$AB = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 9 & 10 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} (4)(9)+(2)(2) & (4)(10)+(2)(5) \\ (3)(9)+(1)(2) & (3)(10)+(1)(5) \end{bmatrix}$$

2×2 2×2

check number of cols in A equals number of rows in B.

$$= \begin{bmatrix} 40 & 50 \\ 29 & 35 \end{bmatrix}$$

over →

Put it all together...

$$(-2B^T + AB) = \begin{bmatrix} -18 & -4 \\ -20 & -10 \end{bmatrix} + \begin{bmatrix} 40 & 50 \\ 29 & 35 \end{bmatrix}$$

$$= \begin{bmatrix} 22 & 46 \\ 9 & 25 \end{bmatrix}$$

iii) inverse of B – inverse of 2x2 matrix

Need: $\det B = ad - bc$ $\quad B^{-1} = \frac{1}{\det B}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$$\det B = (9)(5) - (10)(2)$$
$$= 25$$

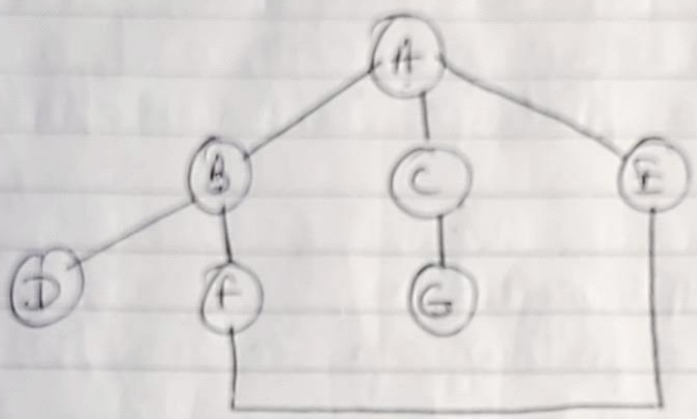
i.e. $B = \begin{matrix} a & b \\ \end{matrix}$
$$\begin{bmatrix} 9 & 10 \\ 2 & 5 \\ c & d \end{bmatrix}$$

$$\therefore B^{-1} = \frac{1}{25}\begin{bmatrix} 5 & -10 \\ -2 & 9 \end{bmatrix} = \begin{bmatrix} \frac{5}{25} & \frac{-10}{25} \\ \frac{-2}{25} & \frac{9}{25} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{5} & \frac{-2}{5} \\ \frac{-2}{25} & \frac{9}{25} \end{bmatrix}$$

simplify fractions where possible

b) See Notes for description of each graph traversal method.



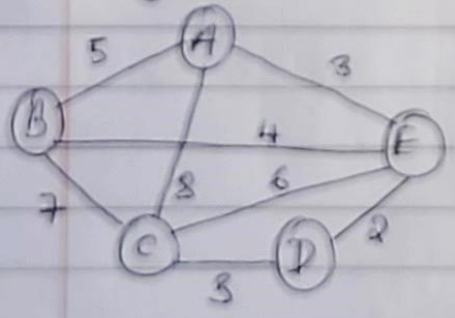
Question states to start at node C!

Breadth-first Search (BFS) uses a queue data structure to help keep track in the algorithm.

Queue: ~~A~~ ~~G~~ ~~B~~ ~~E~~ ~~D~~ ~~F~~

BFS output: C A G B E D F

c) Minimal Spanning Tree (MST) using Prim's algorithm + give the total weight OF MST



T = { A, ae E, ed D, dc C, eb B }

↓ Do not have to go from last vertex added, look at all vertices currently in T

total weight = 3+2+3+4 = 12